CJC
Complete Java Classes

SUPER AND THIS KEYWORD TASK: -

Question 1. Write difference between super () and this () on your theory notebook?

Question 2. Check the program correct or not. If yes what is the output or if not why?

```
public class Parent
{
    public Parent()
    {
     System.out.println("constructor_parent");
      }
   public Parent(String s)

   {

   this();
  System.out.println("parameterized constructor_parent");
 }
}
```

```
public class Child extends Parent
{
public Child()
   {
    super("Hi");
   System.out.println("Constructor_child");
   }
   public static void main(String args[])
   {
    Child c= new Child();
   }
}
```

Question 3. Check the program correct or not. If yes what is the output or If not why?

```
public class A {
  public  A()
  {
    System.out.println("No-arg-constructor_A");
    }
public A(int x)
{
 super();
     System.out.println("arg-constructor_A");
    }
}
```

```
public class B extends A {
  public  B()
  {
    System.out.println("No-arg-constructor_B");
    this("hi");
}
public B(String x)
{
 super();
    System.out.println("arg-constructor_B");
    this();
}
public static void main(String[] args)
{
B b=new B();
    B b1=new B("cjc");
     }
}
```

Question 4. Check the output of program

```
public class A
{

    public A()
    {
    this(10.1f);

     System.out.println("No-arg-constructor_A");

  }

public A(int x){

this();

 System.out.println("int_arg-constructor_A");

  }

public A(double x)

{

     System.out.println("double_arg- constructor_A");

    }

}



public class B extends A
 {
  public B()

  {

     System.out.println("No-arg-constructor_B");

  }
```

```
public B(String x)

{

 super(10);
    System.out.println("arg-constructor_B");
  }


public static void main(String[] args)
{
 B b=new B();
     B b1=new B("cjc");
   }
}
```

Question 5. What is the output of program.

```
 public class Person {
     public void msg(){
             System.out.println("person class");
     }
 }
 public class Student extends Person {
     public void msg()

     {

      System.out.println("student class");
     }
```

```
public void display()
{
msg();
        super.msg();
}
}
```

```
public class Test
{
    public static void main(String args[])
    {
     Student s = new Student();
    s.display();
    }
}
```

Question 6. What is the output of program.

```
public class A{
    String color="white";
    public A(){
            System.out.println("In A Constructor");
     }
}
```

```
public class B extends A {

    String color="black"; //green

    public void display()

    {

        String color="green";

   System.out.println(color);

   System.out.println(this.color);

   System.out.println(super.color);

    }

}


public class Test {

   public static void main(String args[])

   {

    B b=new B();

    b.display();
   }

}
```

Question 7. What is the output of program.

```
public class Animal {

     int cat=10;
```

```
public void eat()
{
System.out.println("cat= "+cat);
System.out.println("eating...");
}
public void eat(String msg)
{
int cat=30;
System.out.println("cat= "+cat);
System.out.println("eating..."+msg);
this.eat(10);
}

public void eat(long l)
{
System.out.println("cat= "+cat);
System.out.println("eating...value:          "+l);
}
}
```

public class Dog extends Animal{

```
public void eat()
{
System.out.println("cat= "+super.cat);
System.out.println("eating  bread...");
}
```

```
public void bark()
{
        System.out.println("barking...");
 }
public void work(){
      super.eat("Mango");
      eat();
      bark();
}
public static void main(String args[]){

 Dog d=new Dog();
      d.work();
}
}
```

Question 8. Check the output of the program.

```
public class Person {

    int id;

    String name;


        public Person(int id, String name)
        {
         this.id=id;
            this.name=name;

      }

    }
    public class Employee extends Person {

     double salary;

     public Employee (int id, String name,double salary)

     {

     super(id,  name);

     this.salary=salary;
     }

     public void display()
     {

     System.out.println("Id: "+id+" Name: "+name+" Salary: "+salary);

       }

     }
```

```
public class Test
{

public static void main(String[] args){

        Employee e1=new Employee(1,"abc",45000);

        e1.display();
      }

 }
```

Question 9. What will the output.

```
     public class Parent {
public Parent(){
          System.out.println("constructor_parent");

   }

   public Parent(String s){

          System.out.println("parameterized constructor_parent");

   }

  }


   public class Child extends Parent
    {

       public Child(){

          this(10);
              System.out.println("Constructor_child");

      }
```

```
public void msg()
{
  System.out.println("msg_method_child");

}

public void display(){
     msg();
}

public Child(float x){
     super();
     System.out.println("float_Constructor_child");

}

public static void main(String args[]){

 Child c= new Child();
     c.display();

  }

}
```

Question 10. What will be the output of the following program?

```
public class Animal{

  int i = 51;

  public Animal(int j){
        System.out.println(i);
        this.i = j * 10;
   }

}
```

```java
public class Dog extends Animal{
   public Dog (int j)
   {
    super(j);
   System.out.println(i);
   this.i = j * 20;
   }
   }
public class Test{
   public static  void main(String[] args){
   Dog d = new Dog (26);
            System.out.println(d.i);
   }
}
```

THANK YOU